



IILM UNIVERSITY, GREATER NOIDA



INTERNSHIP REPORT PRESENTATION

Ethical Hacking

Virtual Internship

AICTE – EduSkills Virtual Internship Program

Tarun Kumar

Roll No. 2410030767 | B.Tech CSE

June – August 2026



**Tarun
Kumar**

B.Tech – Computer Science & Engineering

ABOUT THE STUDENT

Candidate Profile



Roll Number
2410030767



Institute
IILM University, Greater Noida, U.P.



Programme
B.Tech CSE, Batch 2024–2028



Internship Domain
Ethical Hacking



Duration
8 Weeks | June – August 2026

ABOUT THE PROGRAM

EduSkills & AICTE

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under the theme “Nation Building Through Skills.” It partners with the All India Council for Technical Education (AICTE) to run the AICTE–EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal.



ISO Certified

ISO 9001:2015 & ISO/IEC 20000-1:2018
certified organization.



AICTE Patronage

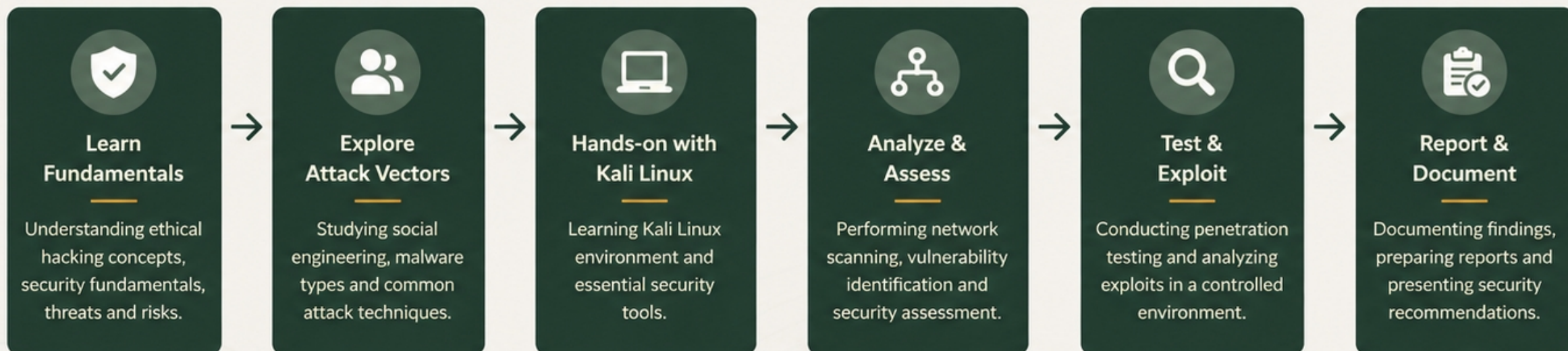
Run under the AICTE National
Internship Portal.

PROBLEM STATEMENT

Bridging Theory and Practical Ethical Hacking

While many students learn cybersecurity concepts theoretically, very few get structured, hands-on exposure to real-world security testing and vulnerability assessment using industry-standard tools like Kali Linux. Topics such as network scanning, vulnerability identification, and penetration testing are often understood only conceptually — with limited practical experience in performing and documenting security assessments.

This internship addressed that gap — through a structured 8-week learning journey covering ethical hacking concepts and practical security testing:



PROJECT OBJECTIVES

What the Internship Set Out to Achieve

The 8-week Ethical Hacking internship was designed to build a strong foundation in cybersecurity concepts and hands-on skills for identifying, analyzing, and addressing security vulnerabilities.



1. Introduction to Ethical Hacking

Week 1

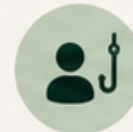
Understand the purpose, scope, and legal aspects of ethical hacking. Explore tools, techniques, and career opportunities.



2. Fundamentals of Security

Week 2

Learn core cybersecurity principles, types of threats, vulnerabilities, and best security practices.



3. Social Engineering

Week 3

Study social engineering attacks like phishing and impersonation and learn prevention techniques.



4. Malware and Their Types

Week 4

Explore different types of malware, their behavior, impact, and mitigation strategies.



5. Networking Essentials

Week 5

Understand networking concepts, protocols, IP addressing, and network communication.



6. Introduction to Kali Linux

Week 6

Get familiar with Kali Linux environment and essential tools used for ethical hacking.



7. Web Application & Capstone Project

Week 7

Learn common web vulnerabilities and perform vulnerability assessment for the capstone project.



8. Penetration Testing Using Kali Linux

Week 8

Perform hands-on penetration testing, exploit vulnerabilities, and prepare security reports.

TECHNOLOGIES & TOOLS

The Toolkit Behind the Internship



Kali Linux

Primary platform for ethical hacking and security testing.



Network Scanning Tools

Tools like Nmap for network discovery and port scanning.



Vulnerability Assessment Tools

Identifying security weaknesses using industry-standard tools.



Penetration Testing Tools

Exploiting vulnerabilities and analyzing system security.



Web Application Testing Tools

Testing web applications for common vulnerabilities.



Password Attacking Tools

Assessing password strength and login security.



Wireless Testing Tools

Analyzing wireless networks for security issues.



Reporting & Documentation

Documenting findings and preparing security reports.

From Learning to Real-World Security Impact



1. Learning Layer

Gained knowledge of cybersecurity fundamentals, ethical hacking concepts, and industry best practices.



2. Tools & Environment Layer

Used industry-standard tools and Kali Linux environment for hands-on security testing and analysis.



3. Assessment Layer

Performed network scanning, vulnerability assessment, and enumeration to identify security weaknesses.



4. Exploitation Layer

Exploited vulnerabilities in controlled labs and capture the flags/credentials to understand real attack scenarios.



5. Analysis & Reporting Layer

Analyzed results, documented findings, and prepared security reports with remediation recommendations.



6. Real-World Impact Layer

Applied learned skills to real-world cybersecurity challenges and strengthened overall security awareness.

METHODOLOGY

The 8-Week Learning Journey

1 WEEK 1



Introduction to Ethical Hacking

- Understanding ethical hacking, its scope, and career opportunities
- Overview of cybersecurity fundamentals and lab setup

2 WEEK 2



Fundamentals of Security

- Learn types of threats, vulnerabilities, and security principles
- Study OSI model, TCP/IP, and common attack vectors

3 WEEK 3



Social Engineering & Reconnaissance

- Learn social engineering techniques like phishing, pretexting, baiting
- Perform information gathering using footprinting & OSINT tools

4 WEEK 4



Network Scanning & Enumeration

- Perform network scanning using Nmap and host discovery
- Enumerate services, ports, and operating systems

5 WEEK 5



Vulnerability Assessment

- Identify and analyze vulnerabilities using industry-standard tools
- Understand CVE, CVSS scoring, and risk assessment

6 WEEK 6



Exploitation & Gaining Access

- Exploit vulnerabilities in a controlled lab environment
- Learn privilege escalation and maintaining access

7 WEEK 7



Post-Exploitation & Reporting

- Perform post-exploitation activities and data extraction
- Document findings and prepare professional security reports

8 WEEK 8



Capstone Project & Review

- Work on a real-world penetration testing scenario
- Final report presentation and overall learning review

EXPECTED OUTCOMES



Skills & Capabilities Gained



Strong Foundation in Cybersecurity: Gained in-depth understanding of ethical hacking concepts, methodologies, and industry best practices.



Hands-on Proficiency with Tools: Worked extensively with industry-standard tools like Kali Linux, Nmap, Wireshark, Burp Suite, Metasploit, and Gobuster for security testing and analysis.



Network & System Analysis: Learned network scanning, enumeration, and vulnerability assessment techniques to identify security weaknesses in systems and applications.



Exploitation Skills: Gained practical experience in exploiting vulnerabilities in controlled environments and understanding attack vectors to strengthen defensive strategies.



Web Application Security: Performed web application testing including OWASP Top 10 vulnerabilities such as SQL Injection, XSS, CSRF, and Command Injection.



Post-Exploitation & Reporting: Executed post-exploitation techniques, privilege escalation, and documented findings with detailed security reports and remediation recommendations.



Professional Growth: Enhanced problem-solving, analytical thinking, and ethical mindset while preparing for real-world cybersecurity challenges.

CERTIFICATION

Successfully Completed



Grade

O – Outstanding



Certificate ID

2026-0BAC011ABD



Duration

8 Weeks | June – August 2026



CERTIFICATE OF INTERNSHIP



EduSkills
ACADEMY

PRESENTED TO

Tarun Kumar

has successfully achieved the Internship Credential by completing an 8-week program in the domain of

Ethical Hacking

During the internship, the learner demonstrated proficiency in:

- Introduction To Ethical Hacking
- Fundamentals Of Security & Social Engineering and Human Security Risks
- Malware: Types, Behaviour, and Impact
- Network Essentials: Foundations of Secure Communication
- Kali Linux Tools and Security Testing Basics
- Basic web application (Vulnerability Assessment)
- Basic Web application penetration testing
- Capstone Project

This certification recognizes the successful completion of the internship requirements.



Issued on: Jul 28, 2026

Certificate ID: 2026-0BAC011ABD

Mitu Swain
General Manager
(Learning & Development)
EduSkills



Thank You

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